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Advance Java

DATABASE Notes

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Company Name: Smart Programming (+91 62838-30308)

Address : Chandigarh & Mohali (Punjab), India

Websites: https://www.smartprogramming.in/

https://courses.smartprogramming.in

Android App:
https://play.google.com/store/apps/details?id=com.smartprogramming

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What is Data ?

Data is collection of the information and they are contain the image, video, audio, text, number etc..

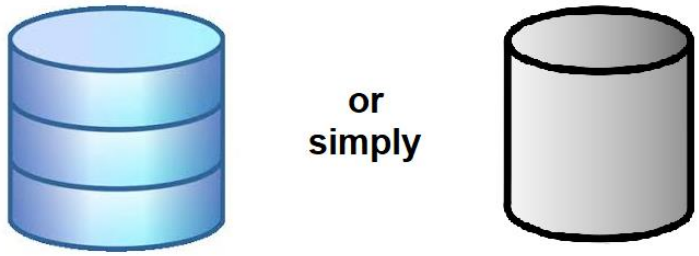
- Data is unorganised raw facts/material that needs to be processed without which it is seemingly random and not that much helpful to humans
- Data needs to be processed in order to get information
- There are multiple types of data forms like single character, boolean, text (string), number (integer or floating-point etc) or picture or sound or video etc.
- In a computer's storage, data is a series of bits (binary digits) that have the value one or zero.

Difference between Data & Information

Basis for Comparison	Data	Information
Meaning	Data means raw facts gathered about someone or something, which is bare and random.	Facts, concerning a particular event or subject, which are refined by processing is called information.
What is it?	It is just text and numbers.	It is refined data.
Based on	Records and Observations	Analysis
Form	Unorganized	Organized
Useful	May or may not be useful.	Always
Specific	No	Yes
Dependency	Does not depend on information.	Without data, information cannot be processed.

What is Database ?

- A *database* is an organized collection of structured information or data, typically stored electronically in a computer system.
- A *database* is usually controlled by a *database* management system (DBMS)
- The **main purpose** of the database is to operate a large amount of information by storing, retrieving, and managing data.
- There are many **databases available** like MySQL, Sybase, Oracle, MongoDB, Informix, PostgreSQL, SQL Server, etc.
- A cylindrical structure is used to display the image of a database.



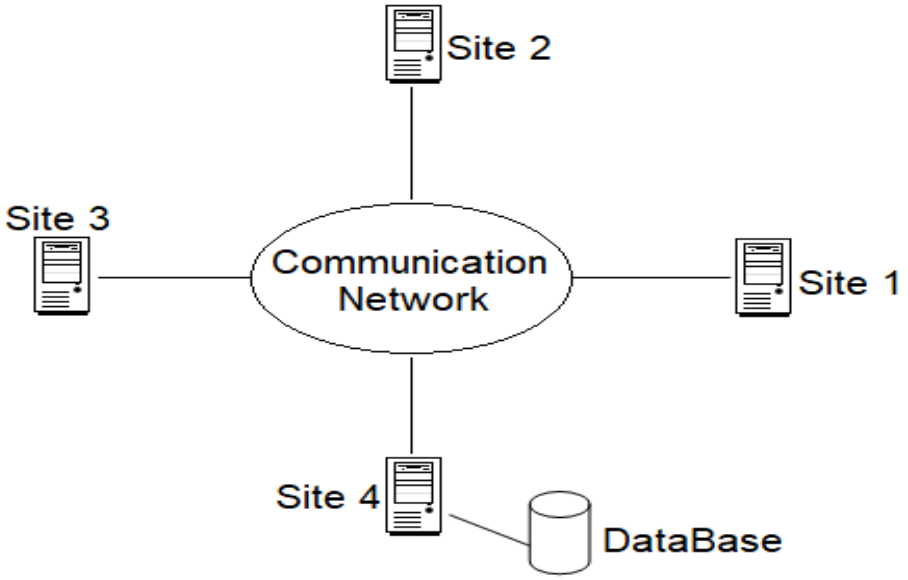
Types of Database

1. Centralized Database
2. Distributed Database
3. Relational Database
4. NoSQL Database
   a. Key-Value storage Database
   b. Document Database
   c. Graph Database
   d. Column Oriented Database
5. Cloud Database
6. Object Oriented Database

7. Hierarchical Database
8. Network Database
9. Personal Database
10. Operational Database

1. Centralized Database

- A centralized database is a database that is located, stored, and maintained in a single location.
- This location is most often a central computer or database system, for example a desktop or server CPU, or a mainframe computer.
- A centralized database is usually accessed using an internet connection such as a LAN or WAN.
- The centralized database is used by organisations such as colleges, companies, banks etc.
- Below is the diagram to show centralized database.



Advantages of Centralized Database

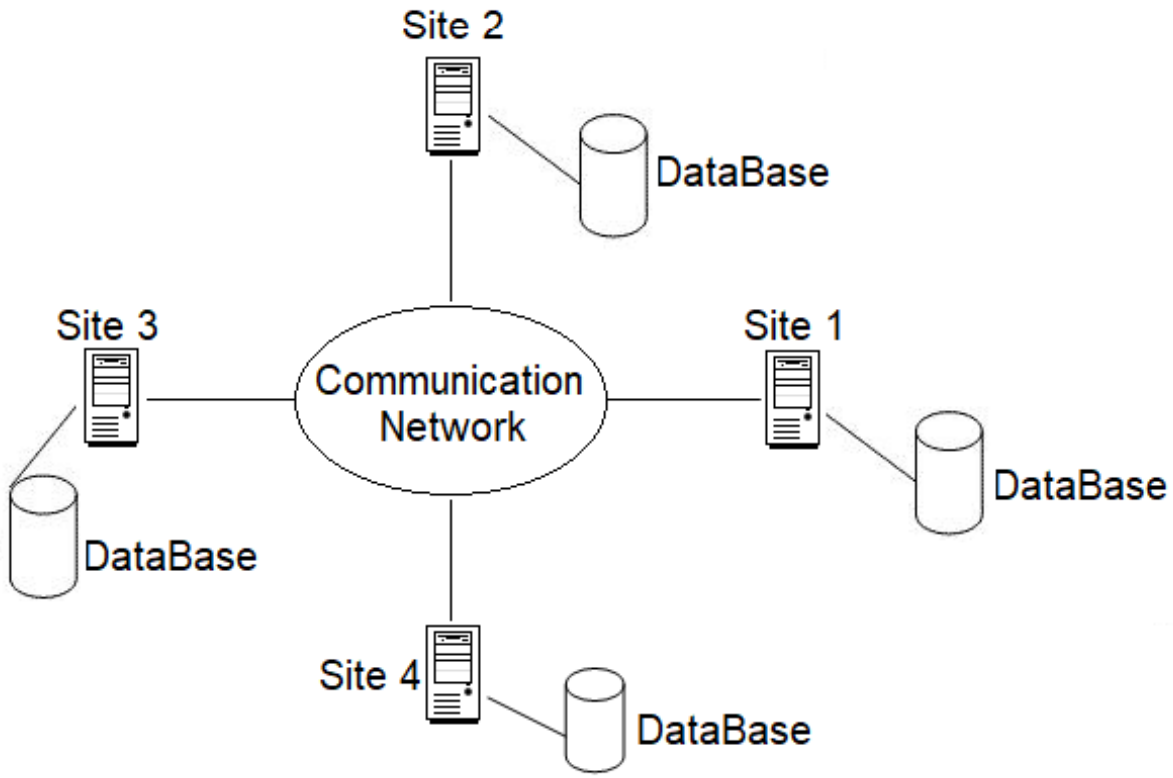
1. The data integrity is maximized as the whole database is stored at a single physical location. This means that it is easier to coordinate the data and it is as accurate and consistent as possible.
2. The data redundancy is minimal in the centralized database
3. Since all the data is in one place there can be stronger security measures around it so the centralized database is much more secure.
4. Data is easily portable because it is stored at the same place
5. The centralized database is cheaper than other types of databases as it required less human power and maintenance
6. All the information in the centralized database can be easily accessed from the same location and at the same time

Disadvantages of Centralized Database

1. Since all the data is at one location, it takes more time to search and access it. If the network is slow, this process takes even more time.
2. There is a lot of data access traffic for the centralized database. This may create a bottleneck situation
3. Since all the data is at same location, if multiple users try to access it simultaneously, it creates a problem. This may reduce the efficiency of the system
4. If there are no database recovery measures in place and a system failure occurs, then all the data in the database will be destroyed.

2. Distributed Database

- A distributed database is a database in which data is stored across different physical locations.
- It may be stored in multiple computers located in the same physical location; or maybe dispersed over a network of interconnected computers.
- Below is the diagram for distributed database



Advantages of DDBMS

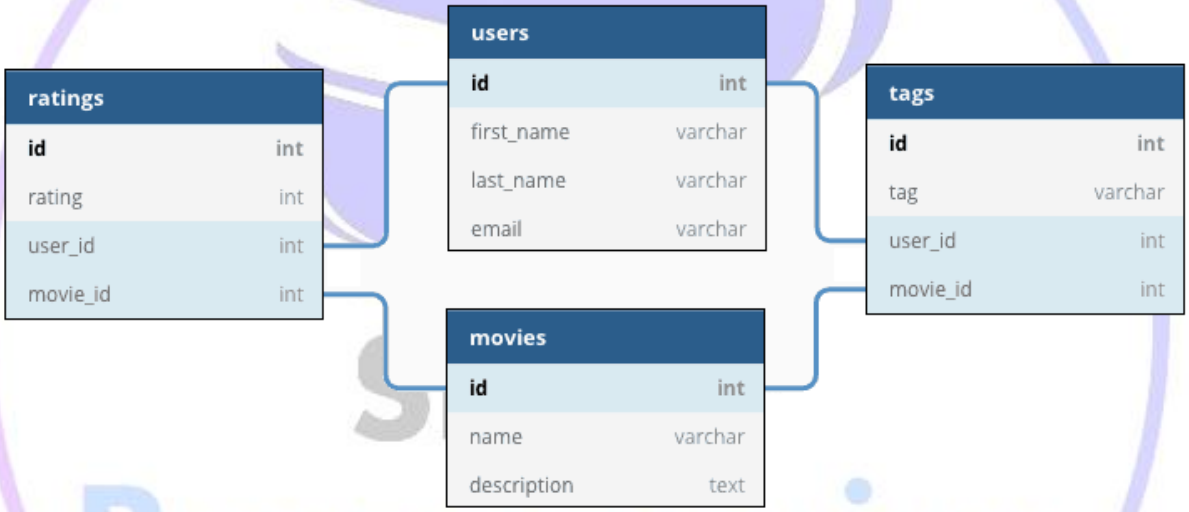
- The database is easier to expand as it is already spread across multiple systems and it is not too complicated to add a system.
- The distributed database can have the data arranged according to different levels of transparency i.e data with different transparency levels can be stored at different locations.
- The database can be stored according to the departmental information in an organisation. In that case, it is easier for a organisational hierarchical access.
- there were a natural catastrophe such as fire or an earthquake all the data would not be destroyed it is stored at different locations.
- It is cheaper to create a network of systems containing a part of the database. This database can also be easily increased or decreased.
- Even if some of the data nodes go offline, the rest of the database can continue its normal functions.

Disadvantages of DDBMS

- The distributed database is quite complex and it is difficult to make sure that a user gets a uniform view of the database because it is spread across multiple locations.
- This database is more expensive as it is complex and hence, difficult to maintain.
- It is difficult to provide security in a distributed database as the database needs to be secured at all the locations it is stored. Moreover, the infrastructure connecting all the nodes in a distributed database also needs to be secured.
- It is difficult to maintain data integrity in the distributed database because of its nature. There can also be data redundancy in the database as it is stored at multiple locations.
- The distributed database is complicated and it is difficult to find people with the necessary experience who can manage and maintain it.

3. Relational Database

- E.F. Codd invented the database in 1970.
- Relational database is based on the relational data model, which stores data in the form of rows(tuple) and columns(attributes), and together forms a table(relation).
- A relational database uses SQL for storing, manipulating, as well as maintaining the data. Each table in the database carries a key that makes the data unique from others.



- **Examples** of Relational databases are MySQL, Microsoft SQL Server, Oracle, PostgreSQL, DB2 etc.

Properties of Relational Database

There are following four commonly known properties of a relational model known as ACID properties, where:

- **A means Atomicity:** This ensures the data operation will complete either with success or with failure. It follows the 'all or nothing

strategy. For example, a transaction will either be committed or will abort.

- **C means Consistency:** If we perform any operation over the data, its value before and after the operation should be preserved. For example, the account balance before and after the transaction should be correct, i.e., it should remain conserved.
- **I means Isolation:** There can be concurrent users for accessing data at the same time from the database. Thus, isolation between the data should remain isolated. For example, when multiple transactions occur at the same time, one transaction effects should not be visible to the other transactions in the database.
- **D means Durability:** It ensures that once it completes the operation and commits the data, data changes should remain permanent.

Advantages of relational database :-

1. Prevents Data redundancy: In relational database management systems the tables with the specific data have relation among them that's why the required data is taken from the previous tables which prevents the data redundancy.
2. Data security: In a relational database management system the data access is privileged which means that the database administrator has the authority of giving access of data to some particular users which makes the data secure.
3. Easy to use: This type of database uses tables which is better and easy to create and use.
4. **Data independence**: The structure of a database can be changed without having to change any application.
5. **Scalable**: Regarding a number of records, or rows, and the number of fields, a database should be enlarged to enhance its usability.

Disadvantages of relational database :-

- Few relational databases have limits on field lengths which can't be exceeded.
- Relational databases can sometimes become complex as the amount of data grows, and the relations between pieces of data become more complicated.
- Complex relational database systems may lead to isolated databases where the information cannot be shared from one system to another.

4. NoSQL Database:

- Non-Relational database store data without structure mechanism to link data from different tables to one another
- Requires low cost hardware
- Much faster performance (read/write) compared to relational database
- Horizontal scaling (adding new server in existing server) is possible
- Never provide table with flat fixed column records (means for eg. every person will have same data like name email password etc). It means schema-free
- Best suited for online analytical processing (data-ware house) Eg. MongoDB, Cassandra, DynamoDB (Amazon), postgreSQL

- Types of NoSQL database :-
    1. Column Database – Cassandra, Hbase (stores in form of column)
    2. Document Database – MongoDB, CouchDB, RavenDB (eg posting image on fb with status, tags etc)
    3. Key-Value Databases – Redia, Riak, DynamoDB, Tokyo Cabinet
    4. Graph Based Database – Neo4J, FlockDB

1. **Coloumn Database** – It is a database that stores the data in columns instead of rows
   In a columnar DB, all the column 1 values are physically together followed by all the column 2 values

2. **Document Database :-** It makes easier for developers to store and querying data in a db by using the same document model format which they use in their application code
   Document db's are efficient for storing catalogues
   Store semi-structure data as document typically in JSON or XML format

3. **Key-Value DB** – It is a software DB that uses an associative array (similar to dictionary) as a fundamental model where each key is associated with one and only one value in collection
   It allows horizontal scaling
   Used cases – Shopping cart and session store in app like FB and twitter
   They improve application performance by storing critical pieces of data in memory for low latency access

4. **Graph based database** :- It is the collection of nodes and edges. Each node represented an entity (like person) and each edge represent a connection or relationship between two nodes.

5. Cloud Database

- A cloud database is a database that typically runs on a cloud computing platform, and access to the database is provided as-a-service.
- Database services take care of scalability and high availability of the database.
- Amazon Web Service, Google Cloud, Google Drive, One Drive, Microsoft Azure etc

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What is DBMS ?

- **Database Management System (DBMS)** is a software for storing and retrieving users' data while considering appropriate security measures. It consists of a group of programs which manipulate the database.
- The DBMS accepts the request for data from an application and instructs the operating system to provide the specific data. In large systems, a DBMS helps users and other third-party software to store and retrieve data.

Characteristics of Database Management System

- Provides security and removes redundancy
- Self-describing nature of a database system
- Insulation between programs and data abstraction
- Support of multiple views of the data
- Sharing of data and multiuser transaction processing
- DBMS allows entities and relations among them to form tables.
- It follows the ACID concept (Atomicity, Consistency, Isolation, and Durability).
- DBMS supports multi-user environment that allows users to access and manipulate data in parallel.

DBMS vs. Flat File

DBMS	Flat File Management System
Multi-user access	It does not support multi-user access
Design to fulfill the need for small and large businesses	It is only limited to smaller DBMS system.

Remove redundancy and Integrity	Redundancy and Integrity issues
Expensive. But in the long term Total Cost of Ownership is cheap	It's cheaper
Easy to implement complicated transactions	No support for complicated transactions

Popular DBMS Software

Here, is the list of some popular DBMS system:

- MySQL
- Microsoft Access
- Oracle
- PostgreSQL
- dBASE
- FoxPro
- SQLite
- IBM DB2
- LibreOffice Base
- MariaDB
- Microsoft SQL Server etc.

Application of DBMS

Sector	Use of DBMS
Banking	For customer information, account activities, payments, deposits, loans, etc.
Airlines	For reservations and schedule

	information.
Universities	For student information, course registrations, colleges and grades.
Telecommunication	It helps to keep call records, monthly bills, maintaining balances, etc.
Finance	For storing information about stock, sales, and purchases of financial instruments like stocks and bonds.
Sales	Use for storing customer, product & sales information.
Manufacturing	It is used for the management of supply chain and for tracking production of items. Inventories status in warehouses.
HR Management	For information about employees, salaries, payroll, deduction, generation of paychecks, etc.

Types of DBMS

Four Types of DBMS systems are:

- Hierarchical database
- Network database
- Relational database
- Object-Oriented database

Hierarchical DBMS

In a Hierarchical database, model data is organized in a tree-like structure. Data is Stored Hierarchically (top down or bottom up) format. Data is represented using a parent-child relationship. In Hierarchical DBMS parent may have many children, but children have only one parent.

Network Model

The network database model allows each child to have multiple parents. It helps you to address the need to model more complex relationships like as the orders/parts many-to-many relationship. In this model, entities are organized in a graph which can be accessed through several paths.

Relational model

Relational DBMS is the most widely used DBMS model because it is one of the easiest. This model is based on normalizing data in the rows and columns of the tables. Relational model stored in fixed structures and manipulated using SQL.

Object-Oriented Model

In Object-oriented Model data stored in the form of objects. The structure which is called classes which display data within it. It defines a database as a collection of objects which stores both data members values and operations.

Advantages of DBMS

- DBMS offers a variety of techniques to store & retrieve data
- DBMS serves as an efficient handler to balance the needs of multiple applications using the same data
- Uniform administration procedures for data
- Application programmers never exposed to details of data representation and storage.
- A DBMS uses various powerful functions to store and retrieve data efficiently.
- Offers Data Integrity and Security
- The DBMS implies integrity constraints to get a high level of protection against prohibited access to data.
- A DBMS schedules concurrent access to the data in such a manner that only one user can access the same data at a time
- Reduced Application Development Time

Disadvantage of DBMS

DBMS may offer plenty of advantages but, it has certain flaws-

- Cost of Hardware and Software of a DBMS is quite high which increases the budget of your organization.
- Most database management systems are often complex systems, so the training for users to use the DBMS is required.
- In some organizations, all data is integrated into a single database which can be damaged because of electric failure or database is corrupted on the storage media
- Use of the same program at a time by many users sometimes lead to the loss of some data.
- DBMS can't perform sophisticated calculations

RDBMS

- A software system used to maintain relational databases is a **relational database** management system (**RDBMS**).
- R**elational database** systems have an option of using the SQL (Structured Query Language) for querying and maintaining the database.

Difference Between DBMS & RDBMS

DBMS: is a software system that allows Defining, Creation, Querying, Update, and Administration of data stored in data files.

Features:

- Normal book keeping system, Flat files, MS Excel, FoxPRO, XML, etc.
- Less or No provision for: Constraints, Security, ACID rules, users, etc.

RDBMS: is a DBMS that is based on Relational model that stores data in tabular form.

- SQL Server, Sybase, Oracle, MySQL, IBM DB2, MS Access, etc.

Features:

- Database, with Tables having relations maintained by FK
- DDL, DML
- Data Integrity & ACID rules
- Multiple User Access
- Backup & Restore
- Database Administration

The key difference is that RDBMS (relational database management system) applications store data in a tabular form, while DBMS applications store data as files.

Does that mean there are no tables in a DBMS?

There can be, but there will be no "relation" between the tables, like in a RDBMS

No.	DBMS	RDBMS
1)	DBMS applications store **data as file**.	RDBMS applications store **data in a tabular form**.
2)	In DBMS, data is generally stored in either a hierarchical form or a navigational form.	In RDBMS, the tables have an identifier called primary key and the data values are stored in the form of tables.
3)	**Normalization is not** present in DBMS.	**Normalization is** present in RDBMS.
4)	DBMS does **not apply any security** with regards to data manipulation.	RDBMS **defines the integrity constraint** for the purpose of ACID (Atomocity, Consistency, Isolation and Durability) property.
5)	DBMS uses file system to store data, so there will be **no relation between the tables**.	in RDBMS, data values are stored in the form of tables, so a **relationship** between these data values will be stored in the form of a table as well.
6)	DBMS has to provide some uniform methods to access the stored information.	RDBMS system supports a tabular structure of the data and a relationship between them to access the stored information.
7)	DBMS **does not support distributed database**.	RDBMS **supports distributed database**.
8)	DBMS is meant to be for small organization and **deal with small data**. it supports **single user**.	RDBMS is designed to **handle large amount of data**. it supports **multiple users**.
9)	Examples of DBMS are file systems, **xml** etc.	Example of RDBMS are **mysql**, **postgre**, **sql server**, **oracle** etc.

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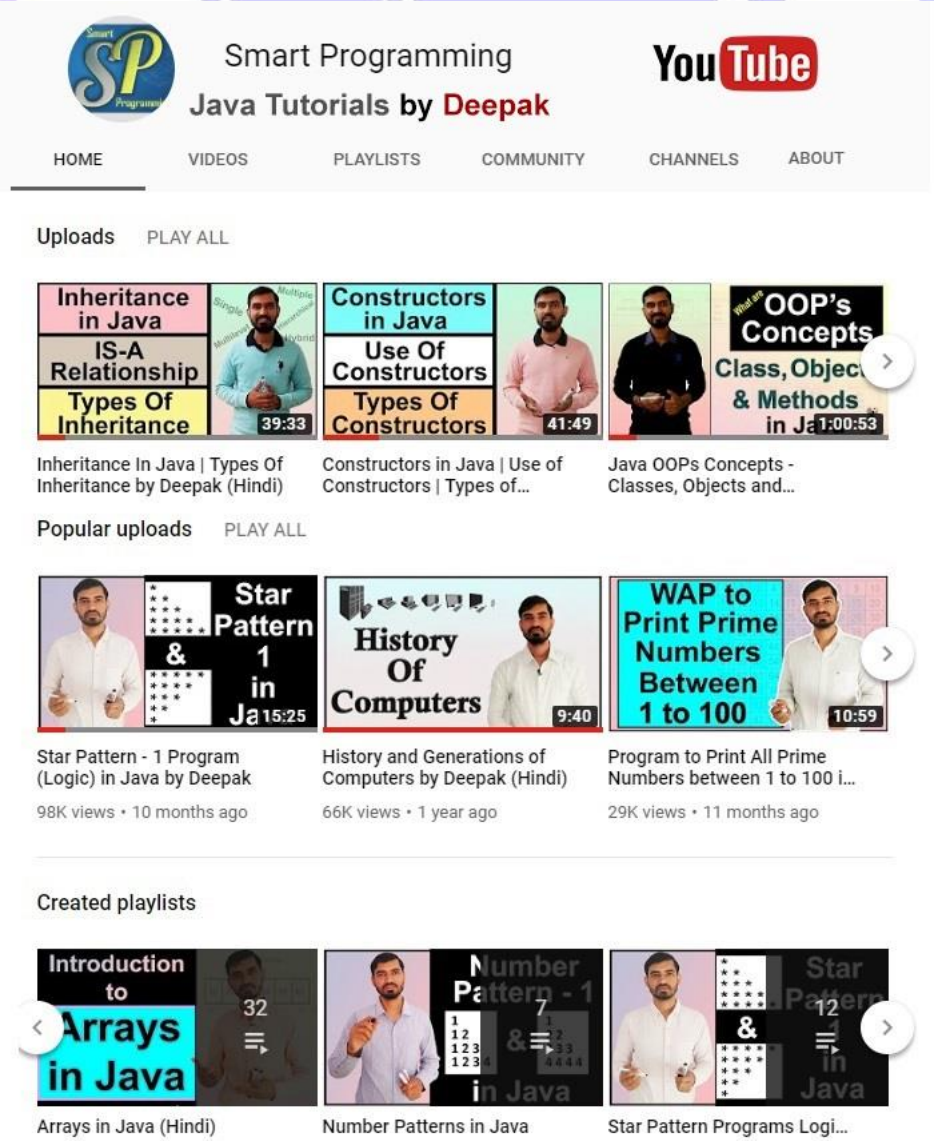
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